Spotted Lantern Fly Terminator

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Objective

The primary objectives of the SLF Terminator project include the development of an innovative bug trap featuring specialized ducts to attract Spotted Lanternflies (SLF). Additionally, the project aims to implement an autonomous servo motor spraying system for targeted pesticide application and a line-tracking movement mechanism for efficient navigation of the rover. These components work together to provide an integrated solution for SLF management.

Executive Summary

The SLF Terminator aims to address the growing threat of the spotted lantern fly (SLF) to American Agriculture and economic stability. The objective is to create an innovative rover equipped with a bug trap, automated insecticide sprayer, camera systems for remote control, and autonomous movement. The group's mission is to substantially reduce the SLF population of Loyola Maryland's Campus, with potential applications in broader agricultural settings.

Risks & Obstacles

Creating the SLF Terminator faced hurdles in team sync, communication, 3D printing delays, and complex programming. Risks included design flaws in vent structures and insecticide sprayer. Overcoming these challenges needed iterative problem-solving, collaboration, and structured design process adherence.

Target Market

The SLF Terminator offers both academic and commercial value through its innovative approach to invasive species control. It combines advanced autonomous technology, precise insecticide systems, and bug tracking, providing real-time data for ecological conservation. For students, this project delivers hands-on experience in CAD, 3D printing, and systems integration, preparing us for real-world engineering careers.

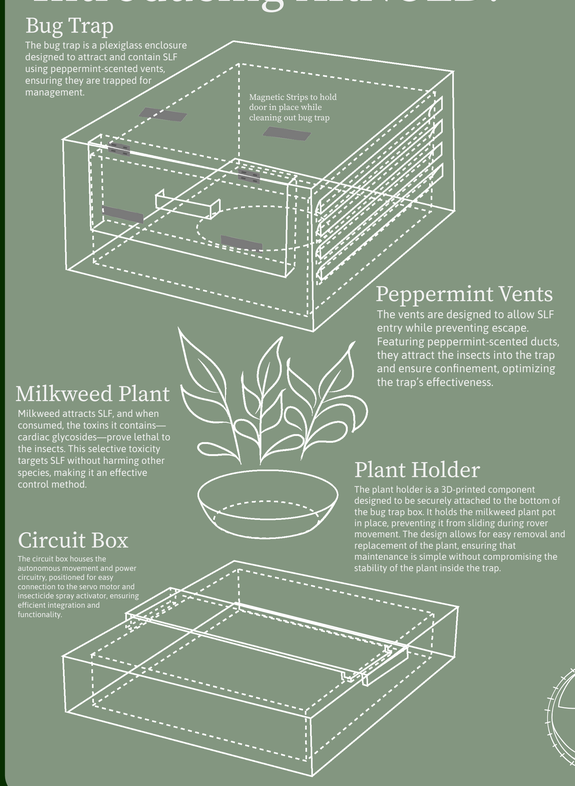
Competitors & Market Research

The SLF Terminator faces competition from traditional manual traps, chemical sprayers, biological control methods, and emerging smart traps. However, it distinguishes itself by combining autonomous movement, precise pesticide spraying, and an innovative bug trap design. This integrated approach offers a scalable, efficient solution for managing the Spotted Lanternfly (SLF) and other invasive species, positioning the SLF Terminator as a unique and environmentally responsible alternative in the pest control market.

Future Work

The SLF Terminator has the potential to autonomously navigate arboretums and farms, mapping high-traffic pest areas for targeted control. Future enhancements could include the ability to count and identify insect species passing through the trap, as milkweed kills only Spotted Lanternflies but not other species. With integrated sensors and cameras, the rover could gather data on local insect populations, ensuring selective pest elimination while preserving biodiversity. This evolution would position the SLF Terminator as a vital tool for invasive species management and ecological monitoring.

Introducing ARNOLD!



The SLF Terminator (ARNOLD) is an autonomous rover designed to combat the Spotted Lanternfly (SLF) infestation. It integrates a plexiglass bug trap with peppermint-scented vents to attract and contain SLF, while a milkweed plant inside delivers lethal toxins to the pests. The rover features a 3D-printed plant holder to keep the milkweed stable yet removable, and a circuit box that houses the rover's power and control systems. Powered by a 12V rechargeable battery, the rover can autonomously navigate and spray insecticide. This design aims to efficiently manage SLF populations in agricultural areas and arboretums while minimizing harm to non-target species.

Project Design Specs

12 V

5.16 Ah

4 hours

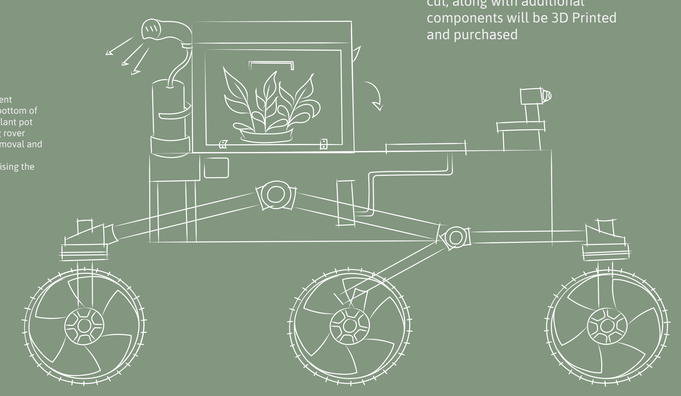
Power will be provided by a rechargable 12V battery that will last our system 4 hours of operation

Controls

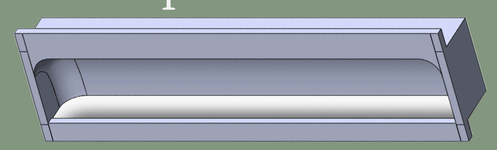
Remote movement, and Autonomous Insecticide spray

Construction

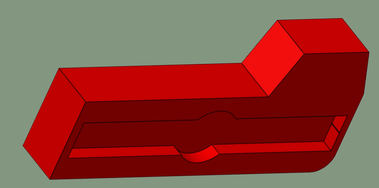
Plexiglass bug trap will be laser cut, along with additional components will be 3D Printed and purchased



CAD Components



This vent design allows Spotted Lanternflies to enter the box while preventing their escape. Due to 3D printer constraints, each vent is half the required length and placed adjacently in the box's front cut-out. Liquid-absorbing ducts, designed to hold peppermint oil, are integrated to attract the lanternflies. The vents are then securely glued to ensure durability.



This design was specifically engineered to attach securely to a servo motor sprocket, ensuring stability. Its primary function is to actuate the sprayer button, toggling its operation. After testing various probe configurations, this version proved the most consistent and reliable.

Conclusion

Interdisciplinary Learning and Skill Development: This project provided hands-on experience in CAD design, 3D printing, electronics, and programming, fostering teamwork and preparing us to tackle real-world engineering challenges with practical, impactful solutions.

Societal and Environmental Impact: The SLF Terminator offers a sustainable, targeted pest control solution to mitigate crop losses and protect livelihoods. Its broader potential includes advancing environmental research and invasive species management, supporting ecosystem preservation and biodiversity.

Thank you!

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Scan here to watch a demonstration of Arnold